Midpoint & Distance Formula — Practice

A worked example comes before each type. Midpoint = Mean, Distance = Difference. Show your thinking.

Name: _____ Date: _____

Type 1 Midpoint on a Number Line

Watch Me — Worked Example

Find the midpoint of -6 and 10.

Midpoint = mean: $\dfrac{-6+10}{2} = \dfrac{4}{2}$. **Answer: 2**

1. Find the midpoint of 4 and 18.

2. Find the midpoint of -9 and 3.

3. Find the midpoint of -7 and -1.

4. Find the midpoint of 2.5 and 8.5.

Type 2 Midpoint in the Coordinate Plane

Watch Me — Worked Example

$A(-2, 5)$ and $B(6, 1)$. Find the midpoint M.

x: $\dfrac{-2+6}{2} = 2$, y: $\dfrac{5+1}{2} = 3$. **Answer: $M(2, 3)$**

5. $(1, 4)$ and $(7, 10)$

6. $(-3, -2)$ and $(5, 6)$

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7. $(0, -5)$ and $(-8, 3)$

8. $(2, 7)$ and $(5, -2)$

Type 3 Find the Other Endpoint

Watch Me — Worked Example

$M(3, 4)$ is the midpoint of $\overline{AB}$, and $A(1, 1)$. Find B.
From A to M: x goes up 2, y goes up 3. Do it again from M: $(3 + 2,\ 4 + 3)$.
(Or: $B = (2 \cdot 3 - 1,\ 2 \cdot 4 - 1)$.) **Answer: $\boldsymbol{B(5, 7)}$**

9. Midpoint $M(4, 6)$, endpoint $A(2, 1)$. Find B.

10. Midpoint $M(0, 0)$, endpoint $A(-3, 5)$. Find B.

11. Midpoint $M(-1, 2)$, endpoint $A(3, 8)$. Find B.

12. On a number line, M is at 7 and A is at 2. Where is B?

Type 4 Distance Formula

Watch Me — Worked Example

$A(1, 1)$ and $B(7, 9)$. Find AB.
Differences: $7 - 1 = 6,\ 9 - 1 = 8.$ $AB = \sqrt{6^2 + 8^2} = \sqrt{100}.$ **Answer: $\mathbf{10}$**

13. $(2, 3)$ and $(5, 7)$

14. $(-1, -2)$ and $(11, 3)$

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15. $(0, 4)$ and $(6, -4)$

16. HARD $(1, 2)$ and $(4, 5)$. Give the exact answer and a decimal.

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